

HW4

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```
# import packages
library(plyr)
library(ggplot2)
library(dplyr)
library(xtable)
library(reshape)
# set seed
set.seed(123)
```

Question 1

Part A

Let $\theta \sim \text{beta}(a, b)$ and let $\psi = \log[\theta/(1 - \theta)]$. Obtain the form of p_ψ and plot p_ψ for the case that $a = b = 1$.

We are given

$$p_\psi(\psi) = p_\theta(h(\psi)) \times \left| \frac{dh}{d\psi} \right|.$$

Let $\psi = g(\theta)$, where g is a monotone function of θ , and h be the inverse of g so that $\theta = h(\psi)$. Since $\theta \sim \text{beta}(a, b)$, we have that the probability density of θ , $p_\theta(\theta)$, is given by

$$p_\theta(\theta) = p_\theta(h(\psi)) = \frac{1}{B(a, b)} * (h(\psi))^{a-1} * (1 - h(\psi))^{b-1}.$$

Since h is the inverse of $g(\theta) = \psi$ and $\psi = \log[\frac{\theta}{1-\theta}]$, it follows that $h(\psi) = \theta = \frac{e^\psi}{e^\psi + 1}$. Observe

$$\begin{aligned}
p_\psi(\psi) &= p_\theta(h(\psi)) \times \left| \frac{dh}{d\psi} \right| \\
&= p_\theta \left(\frac{e^\psi}{e^\psi + 1} \right) \times \left| \frac{d}{d\psi} \left(\frac{e^\psi}{e^\psi + 1} \right) \right| \\
&= \frac{1}{B(a, b)} \times \left(\frac{e^\psi}{e^\psi + 1} \right)^{a-1} \times \left(1 - \frac{e^\psi}{e^\psi + 1} \right)^{b-1} \times \frac{e^\psi}{(e^\psi + 1)^2} \\
&= \frac{1}{B(a, b)} \times \left(\frac{e^\psi}{e^\psi + 1} \right)^{a-1} \times \left(\frac{e^\psi + 1}{e^\psi + 1} - \frac{e^\psi}{e^\psi + 1} \right)^{b-1} \times \frac{e^\psi}{(e^\psi + 1)^2} \\
&= \frac{1}{B(a, b)} \times \left(\frac{e^\psi}{e^\psi + 1} \right)^{a-1} \times \left(\frac{e^\psi + 1 - e^\psi}{e^\psi + 1} \right)^{b-1} \times \frac{e^\psi}{(e^\psi + 1)^2} \\
&= \frac{1}{B(a, b)} \times \frac{e^{\psi(a-1)}}{(e^\psi + 1)^{a-1}} \times \left(\frac{1}{e^\psi + 1} \right)^{b-1} \times \frac{e^\psi}{(e^\psi + 1)^2} \\
&= \frac{1}{B(a, b)} \times \frac{e^{\psi(a-1)} \times e^\psi}{(e^\psi + 1)^{a-1+b-1+2}} \\
&= \frac{1}{B(a, b)} \times \frac{e^{a\psi}}{(e^\psi + 1)^{a+b}}.
\end{aligned}$$

When $a = b = 1$,

$$p_\psi(\psi) = \frac{1}{B((1), (1))} \times \frac{e^{(1)\psi}}{(e^\psi + 1)^{(1)+(1)}} = \boxed{\frac{e^\psi}{(e^\psi + 1)^2}}.$$

Here is the plot of $p(\psi)$ when $a = b = 1$:

```

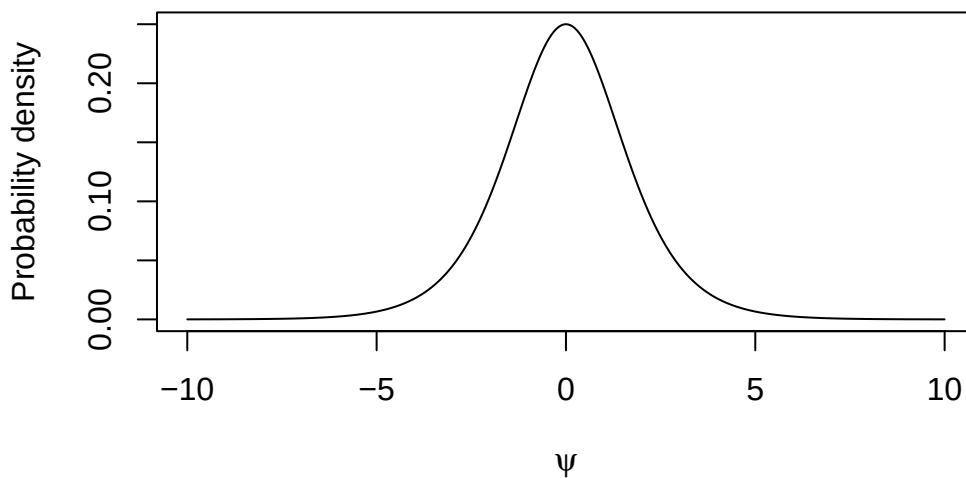
### Plot prob density for psi when a = b = 1 ###
# Create a grid over the support of psi
psi_vals <- seq(-10, 10, length.out = 1000)

# Compute the density values
prob_values <- exp(psi_vals) / (exp(psi_vals) + 1)^2

# Create the plot
plot(psi_vals, prob_values, type = "l", xlab = expression(psi),
     ylab = "Probability density", main = "Density of psi when a = b = 1")

```

Density of psi when a = b = 1



Part B

Let $\theta \sim \text{gamma}(a = \text{shape}, b = \text{rate})$ and let $\psi = \log(\theta)$. Obtain the form of p_ψ and plot p_ψ for the case that $a = b = 1$.

Now, since $\theta \sim \text{gamma}(a = \text{shape}, b = \text{rate})$, we have

$$p_\theta(\theta) = \frac{b^a}{\Gamma(a)} \times \theta^{a-1} \times e^{-b\theta}.$$

Since $\psi = \log(\theta)$, now we have $h(\psi) = \theta = e^\psi$, and $\left| \frac{dh}{d\psi} \right| = e^\psi$. Observe

$$\begin{aligned} p_\psi(\psi) &= p_\theta(h(\psi)) \times \left| \frac{dh}{d\psi} \right| \\ &= p_\theta(e^\psi) \times (e^\psi) \\ &= \frac{b^a}{\Gamma(a)} \times e^{\psi(a-1)} \times e^{-be^\psi} \times e^\psi \\ &= \frac{b^a}{\Gamma(a)} \times e^{a\psi - be^\psi}. \end{aligned}$$

When $a = b = 1$, observe

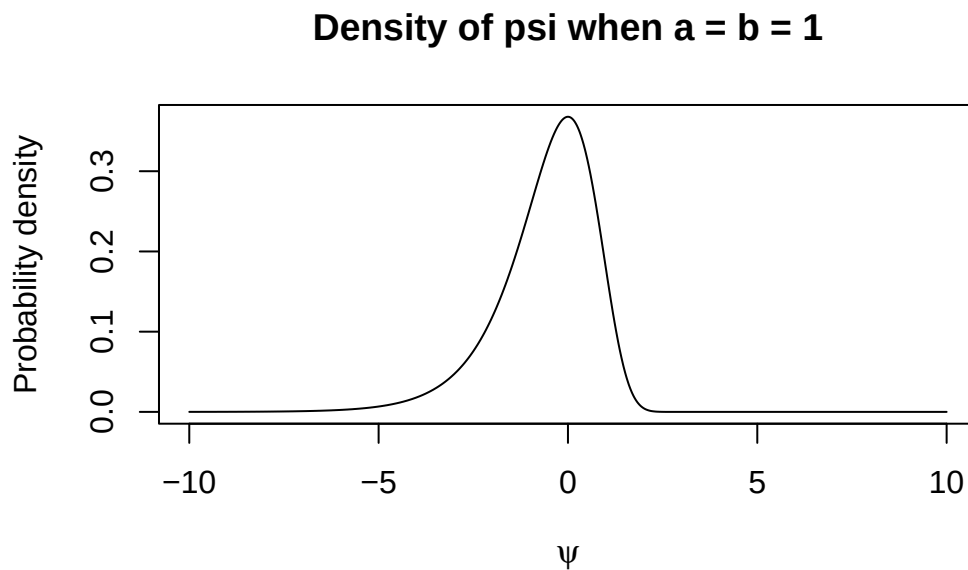
$$p_{\psi}(\psi) = \frac{(1)^{(1)}}{\Gamma((1))} \times e^{(1)\psi - (1)e^{\psi}} = \boxed{e^{\psi - e^{\psi}}}.$$

Plotting gives the following:

```
### Plot prob density for psi when a = b = 1 ###
# Create a grid over the support of psi
psi_vals_new <- seq(-10, 10, length.out = 1000)

# Compute the density values
prob_values_new <- exp(psi_vals - exp(psi_vals))

# Create the plot
plot(psi_vals_new, prob_values_new, type = "l", xlab = expression(psi),
     ylab = "Probability density", main = "Density of psi when a = b = 1")
```



Question 2

Now, we can answer our original question: “What is the posterior probability that $\mu_S > \mu_C$?”

The easiest way to do this is to take a bunch of samples from each of the posteriors, and see what fraction of times we have $\mu_S > \mu_C$. This is an example of a Monte Carlo approximation (much more to come on this in the future).

To do this, we draw $N = 10^6$ samples from each posterior:

Interpret the posterior probability that you compute above.

Lab Task 4

```
## don't use data in package, use data on lab assignment #
# input data
# spurters
x = c(18, 40, 15, 17, 20, 44, 38)
# control group
y = c(-4, 0, -19, 24, 19, 10, 5, 10,
      29, 13, -9, -8, 20, -1, 12, 21,
      -7, 14, 13, 20, 11, 16, 15, 27,
      23, 36, -33, 34, 13, 11, -19, 21,
      6, 25, 30, 22, -28, 15, 26, -1, -2,
      43, 23, 22, 25, 16, 10, 29)
# store data in data frame
iqData = data.frame(Treatment = c(rep("Spurters", length(x)),
                                   rep("Controls", length(y))),
                    Gain = c(x, y))
xLimits = seq(min(iqData$Gain) - (min(iqData$Gain) %% 5),
              max(iqData$Gain) + (max(iqData$Gain) %% 5),
              by = 5)

prior = data.frame(m = 0, c = 1, a = 0.5, b = 50)
findParam = function(prior, data){
  postParam = NULL
  c = prior$c
  m = prior$m
  a = prior$a
  b = prior$b
  n = length(data)
  postParam = data.frame(m = (c*m + n*mean(data))/(c + n),
                          c = c + n,
                          a = a + n/2,
                          b = b + 0.5*(sum((data - mean(data))^2)) +
                            (n*c*(mean(data) - m)^2)/(2*(c+n)))
}
```

```

    return(postParam)
}
postS = findParam(prior, x)
postC = findParam(prior, y)

# sampling from two posteriors

# Number of posterior simulations
sim = 1000000

# initialize vectors to store samples
mus = NULL
lambdas = NULL
muc = NULL
lambdac = NULL

# Following formula from the NormalGamma with
# the update paramaters accounted accounted for below

lambdas = rgamma(sim, shape = postS$a, rate = postS$b)
lambdac = rgamma(sim, shape = postC$a, rate = postC$b)

mus = sapply(sqrt(1/(postS$c*lambdas)),rnorm, n = 1, mean = postS$m)
muc = sapply(sqrt(1/(postC$c*lambdac)),rnorm, n = 1, mean = postC$m)
# task 4

mean(mus>muc) # check if this is right

```

```
[1] 0.970673
```

The posterior probability that the true mean change in IQ score is greater in the spurters group versus the control group is 0.971, meaning that there is a 97.1% chance that the mean change in IQ score is greater in the spurters group versus the control group, given the observed data and chosen prior distributions, based on this approximation of the proportion of posterior draws for which $\mu_S > \mu_C$.

Lab Task 5

Let's return back to the prior assumptions. There are a few ways that you can check that the prior conforms with our prior beliefs. Let's go back and check these. Draw some samples from the prior and look at them—this is probably the best general strategy. See Figure. It's also a good idea to look at sample hypothetical datasets $X_{1:n}$ drawn using these sampled parameter values.

Please replicate a plot similar to Figure 3 and report your findings.

```
# Number of posterior simulations
sim = 1000

# Sample prior precisions
lambda_prior <- rgamma(sim, shape = prior$a, rate = prior$b)

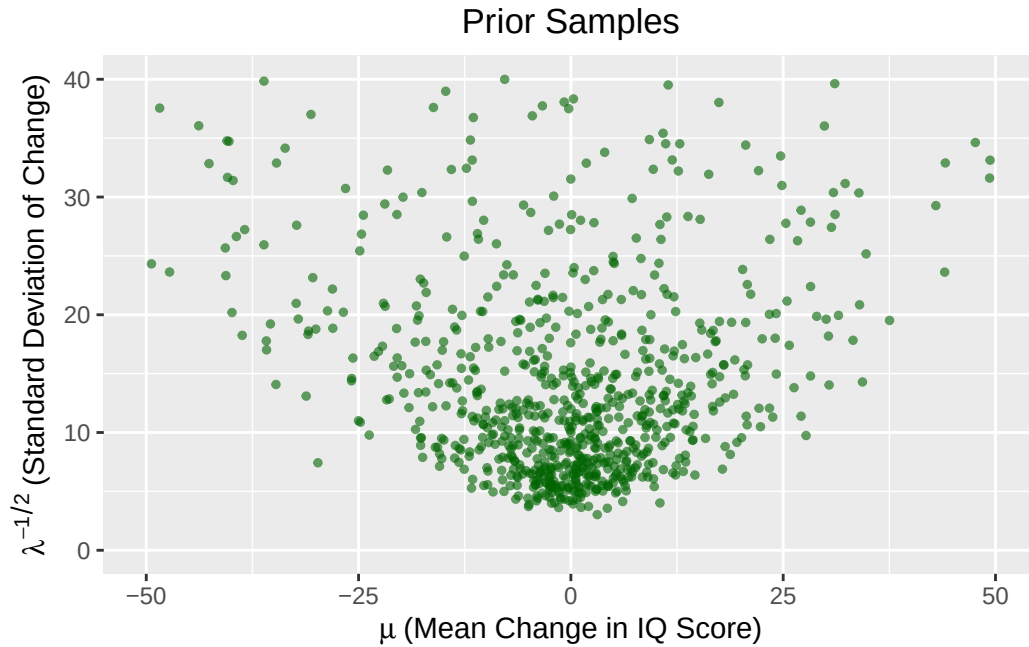
# Sample prior means conditional on precision
mu_prior <- rnorm(sim, mean = prior$m, sd = sqrt(1 / (prior$c * lambda_prior)))

# Convert precision to standard deviation
sd_prior <- lambda_prior^(-0.5)

# Store simulations
simDF_prior <- data.frame(mu = mu_prior, sd = sd_prior)

# Plot prior samples
ggplot(simDF_prior, aes(x = mu, y = sd)) +
  geom_point(color = "darkgreen", alpha = 0.6, size = 1) +
  labs(x = expression(mu~"(Mean Change in IQ Score)"),
       y = expression(lambda~"{-1/2}"~"(Standard Deviation of Change)")) +
  ggtitle("Prior Samples") +
  theme(plot.title = element_text(hjust = 0.5)) +
  xlim(-50, 50) +
  ylim(0, 40)
```

Warning: Removed 212 rows containing missing values or values outside the scale range (`geom\_point()`).



```
# I simplified my code using AI
```

Based on the samples drawn from the scatter plot, we observe that the choice of prior conforms with our prior beliefs. We expect there to be no change in IQ score (i.e., mean is 0) based on our prior beliefs because we do not have any expert information suggesting a change in IQ score. Therefore, we expect most points to be clustered near $\mu = 0$, and we expect relatively large standard deviation to reflect considerable uncertainty about individual variation.

Note: I set limits on the axes to match the sample plot...we may draw samples outside these limits. Here is the same plot without limits below.

```
# Plot prior samples
ggplot(simDF_prior, aes(x = mu, y = sd)) +
  geom_point(color = "darkgreen", alpha = 0.6, size = 1) +
  labs(x = expression(mu~"(Mean Change in IQ Score)"),
       y = expression(lambda^{-1/2}~"(Standard Deviation of Change)")) +
  ggtitle("Prior Samples") +
  theme(plot.title = element_text(hjust = 0.5))
```

